Hyperseed: A Scrutability-Style Core Ontology and a Practical Inference-Control Scaffold for Probabilistic Logic Networks

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Abstract

A recurring aspiration in philosophy, linguistics, and cognitive science is that a wide range of concepts—and perhaps a wide range of truths—can be reconstructed from a compact base vocabulary together with principled inferential bridges. Classical versions include Carnap's constructional systems and Wierzbicka's Natural Semantic Metalanguage; Chalmers' *Constructing the World* reframes the ambition using scrutability: given a compact base and ideal reasoning, many truths are conditionally (and sometimes a priori) scrutable from that base.

We argue that Hyperseed—a formal ontology presented as a semantic stack for mind, experience, and reality—can serve as a workable, engineering-relevant variant of a scrutability base, and that Probabilistic Logic Networks (PLN), embedded within the OpenCog Hyperon architecture, provide a concrete mechanism for building approximate intensional reductions of real-world concepts into Hyperseed terms. Beyond philosophical alignment, the same reductions yield an operational payoff: they can guide PLN inference control via estimates of marginal information gain per unit cost. We outline three complementary control templates (backward chaining, forward chaining, and bidirectional geodesic search) and discuss how domain-anchoring and embodiment links connect Hyperseed abstractions to concrete sensorimotor or domain-specific inference.

Finally, we argue that representing a core ontology explicitly—rather than only implicitly via learned representations—increases a system's capacity for transparent self-understanding and deliberate self-modification, an advantage that becomes increasingly relevant in AGI regimes where reflective control and corrigibility-like properties matter. And we explain how one may treat the current version of Hyperseed as an initial condition for ongoing automated ontology learning, in which an AI system progressively shapes for itself an ontology suitable for optimizing its own operations.

Contents

1 Introduction

1.1 Core ontologies: the philosophical and linguistic background

The idea of defining a compact base vocabulary from which broader conceptual ontologies can be systematically reconstructed has a long lineage. Carnap's *Aufbau* [1] sought constructional systems built from primitive relations; Wierzbicka's Natural Semantic Metalanguage [2] argues for a cross-linguistic inventory of semantic primes; and Chalmers' *Constructing the World* [5] reframes the ambition in terms of scrutability bases and ideal reasoning. Computational resources such as WordNet [3] and FrameNet [4] reflect aspects of these philosophical traditions, though with greatly restricted representational scope.

Despite their diversity, these efforts share two key motifs. First, the base vocabulary need not be tiny, but it should be *compact* in a principled sense. Second, the process of grounding concepts in combinations of core elements must not be trivialized—it is not, in general, as simple as representing a number by its prime factors. For AI, the crucial question is whether such a base can be made *operational*: can it guide inference, learning, and self-understanding rather than merely serving as a philosophical curiosity?

1.2 Why core ontologies still matter for AI

It might seem that the whole approach of ontology-based knowledge representation is a historical atavism, given the success of approaches such as transformer neural networks. Modern AI systems routinely learn powerful latent representations that *behave as if* they contain an ontology: there are stable clusters, latent factors, and compositional features that guide prediction and action. However, these implicit ontologies are encoded in weights, distributed vectors, or opaque activation patterns and are therefore not directly available for explicit reasoning, explanation, or deliberate revision. This makes it difficult for a system to use its own conceptual structure as an object of inference and self-modification.

Classical symbolic AI and knowledge-graph systems have long emphasized explicit ontologies, but their weakness has typically been brittleness, inability to cope with uncertainty, and poor scalability. Probabilistic and nonmonotonic logic frameworks, including Probabilistic Logic Networks (PLN) [7], were developed in part to close this gap: retain explicit conceptual structure while allowing uncertain evidence, graded inference, and practical heuristics for search control. Modern infrastructures such as OpenCog Hyperon [6] provide scalable execution of such frameworks against sizeable knowledge bases, yet scalable inference control remains a challenge. An appropriately constructed ontology can address this challenge by supplying a structured basis for inference-control heuristics—but only if the ontology is broad, deep, and rich enough to support the full subtleties of practical reasoning across a flexible variety of domains.

1.3 Hyperseed, Hyperon, and PLN

The Hyperseed ontology [8] is a collection of over 200 formally defined concepts, together with theorems relating them and establishing their properties. It is organized as a semantic stack spanning mind, experience, and reality, and it includes illustrative theoretical material on topics such as AI ethics and the nature of scientific theory. Crucially, Hyperseed is intended to be *operational*: it should change what a reasoning system *does*, not only what it *says*.

Our proposed approach to leveraging Hyperseed for AGI inference sits within the broader Hyperon initiative [6]—an AGI-oriented framework that treats a knowledge base as a living, incrementally updated structure (a distributed "atomspace" / knowledge store) coupled to multiple inference and learning processes. PLN is Hyperon's uncertain inference engine: it provides a collection of inference rules and truth-value semantics intended to support reasoning with partial, noisy, and conflicting evidence while remaining compositional and scalable. Because PLN inference is combinatorially explosive, practical use requires *inference control*: heuristics and learned policies that decide which rules to apply and which premises to use, under resource budgets.

1.4 Thesis and plan of the paper

We position Hyperseed simultaneously in three roles that are often kept separate:

- **Philosophical role (scrutability / reduction).** Hyperseed serves as an expressive base-language in which other concepts can be *approximately* expressed. PLN provides uncertain, intensional inference for constructing approximate analyses, using (among other things) the Hyperseed principle that "information" is observer-indexed bookkeeping grounded in distinctions and probabilistic belief.[1]

- **Engineering role (inference control).** Hyperseed supplies structure for estimating which inference steps are likely to matter *next*. Inference control is attention-allocation in the reasoning substrate: the management of limited resources[2] bounded by effort budgets.[3] We sketch three control templates—backward, forward, and geodesic—in which Hyperseed guides expected marginal information gain per unit cost and interfaces naturally with association-spreading mechanisms and learned rule-selection priors.

- **AGI / meta-cognitive role (self-understanding and self-modification).** A system whose core ontology is explicit in its deliberative language can reason about, critique, and revise the way its experience is organized. When the ontology is only implicit, the system may still *have* one, but using it for transparent self-understanding and deliberate self-modification becomes substantially harder.

Our central claim is that PLN-based approximate reduction to Hyperseed, implemented in the Hyperon framework, can fulfill a close, workable variant of the desiderata articulated in Chalmers' *Constructing the World* [5]. In the sections that follow, we make this idea formally precise (Section 2), spell out how Hyperseed can guide PLN inference control in several complementary ways (Section 3), identify structural conditions under which ontology-guided inference achieves exponential speedup and illustrate the theory with two worked examples (Section 4), and then argue that making a core ontology explicit is valuable from an AGI perspective (Section 6).

1.5 Automated ontology learning as meta-inference control

Finally, extending the above ideas, we consider the merging of dynamic ontology-formation into the learning process of the AI system using the ontology.

To frame this concept, start with our central premise that an ontology is not merely a naming scheme but an *inference-control scaffold*: by choosing a base vocabulary B (scrutability primitives) and a definitional/bridge theory Δ, we induce reduction profiles and graph-structured relevance relations that shape which inference steps are considered promising under bounded resources. Once we have explicit, quantitative criteria for evaluating this role (e.g. scrutability-based reduction quality, separator/leakage diagnostics, and cost-normalized efficiency measures such as OER), it becomes natural to treat ontology design itself as a learnable optimization problem rather than a purely manual craft.

In this perspective, *automated ontology learning* is a meta-level learning loop that searches over ontologies $\mathcal{O} = (B, \Delta)$ using inference histories as training data. The system runs tasks, records which conceptual reductions and relevance cues actually improved search efficiency, and then proposes ontology edits that would have made those successful traces shorter, more stable, or more reusable. Two complementary routes are especially natural here: (A) *growing an ontology from*

[1]Hyperseed-Concept 98 (*Information as Distinction*) and Hyperseed-Concept 139 (*Probabilistic Belief*) in [8]. These two concepts recur frequently below; subsequent footnote references use abbreviated forms.

[2]Hyperseed-Concept 60 (*Attention*) in [8].

[3]Hyperseed-Concept 100 (*Genenergy / Effort*) in [8].

scratch by inventing new intermediate concepts that compress frequent proof fragments or create better separators in the relevance graph; and (B) *incrementally refining a given scaffold* (e.g. Hyperseed) by targeted edits such as adding mediator concepts, splitting or merging overloaded predicates, and learning/repairing bridge axioms that anchor abstract concepts to domain predicates. In both cases, concepts are treated as logical definitions (not just taxonomy nodes), and candidate edits are accepted only when they improve the same gain-per-cost signals used for ordinary inference control, after accounting for the added representational/maintenance complexity of the ontology itself.

Section 7 sketches concrete algorithms for both routes, including trace-mining candidate concept definitions, replay-based scoring of ontology edits, and conservative budgeted selection of additions/repairs. This meta-loop completes the paper's methodological arc: the same formal quantities that guide *which inference steps* to take can also guide *which conceptual scaffold* the system should build for taking them.

2 PLN-Based Approximate Intensional Reduction to Hyperseed as a Concrete Variant of Chalmers-Style Scrutability

2.1 Motivation

Chalmers [5] argues (in the spirit of Carnap, Russell, and Leibniz) that there is a compact collection of relatively "primitive" expressions (a *scrutability base*) from which all truths are, in an appropriate sense, scrutable. Hyperseed is compatible with this spirit: it is explicitly introduced as an ontology and "core scaffolding" intended to express other concepts, while not claiming to be a final list of semantic primes. Instead, it aims to guide reasoning and enable deep reconstruction via a semantic stack.

The present section makes the Carnap/Chalmers aspiration *algorithmically concrete* by:

1. treating a chosen subset of Hyperseed's ontology as a compact base vocabulary in the Chalmers sense;
2. representing real-world concepts intensionally via properties-with-degrees;
3. using PLN to build and refine approximate reductions of arbitrary concepts to Hyperseed concepts; and
4. controlling this reduction/inference process using an information-theoretic notion of intensional inheritance.

2.2 An AI-friendly "scrutability" relation

Chalmers' scrutability relations capture an *epistemic* notion, often framed in terms of a-priority. In deployed AI systems—and especially in multi-source knowledge graphs—one must work with uncertainty, partial observability, and continual revision.[4] We therefore introduce a *probabilistic* surrogate that keeps the spirit of the Chalmers project while matching the realities of PLN-style inference.

[4] Hyperseed-Concept 139 (*Probabilistic Belief*) and the associated concept *Belief System* in [8].

Definition 1 (Hyperseed base vocabulary)**.** *Let $\mathcal{H}$ denote the set of Hyperseed ontology predicates/relations currently in use by a reasoning system, organized into a small number of semantically coherent families (e.g. patterns, processes, representations, beliefs). A* Hyperseed base vocabulary *is a chosen subset $B \subseteq \mathcal{H}$ intended to be "compact" in the Chalmers sense (few families, no trivial encodings), but still expressive enough to guide reductions and inferences.*

Definition 2 (Probabilistic scrutability from a base)**.** *Fix a probability model $\mathbb{P}$[5] over a space of "cases" Ω (these may be world-histories, situations, or epistemic scenarios indexed by a context).[6] For any propositions S and base-description D, let $S(\omega) \in \{0, 1\}$ denote the truth value of S in case ω.*

*We say that S is ϵ-*scrutable *from the base vocabulary B (relative to a class of allowed inference rules $\mathcal{R}$, e.g. PLN rules) if there exists a formula/derivation $\Phi_B \in \mathrm{Form}(B, \mathcal{R})$ such that*

$$\mathbb{P}(\Phi_B(\omega) \neq S(\omega)) \leq \epsilon.$$

When $\epsilon = 0$ we call this exact *scrutability; otherwise it is* approximate *scrutability.*

Comment. This definition is deliberately close to Chalmers' while relaxing (i) strict a-priority and (ii) strict definitional equivalence. It is designed so that improving the reduction Φ_B directly yields improved inference-control heuristics in an uncertain reasoner.

2.3 Intensional reductions as property profiles

The key gap between a philosophical scrutability thesis and a deployable AI mechanism is that *real concepts* rarely admit crisp necessary-and-sufficient definitions. A workable substitute is to represent a concept by a *profile* of weighted properties: a structured intension rather than an extensional set of instances.

Definition 3 (Intensional property-profile concept)**.** *A concept W is represented by a finite multiset of properties*

$$\mathrm{Prop}(W) := \{(W_1, e_1), \ldots, (W_m, e_m)\}$$

where each W_j is a predicate (often drawn from $\mathcal{H}$) and each $e_j \in [0, 1]$ is a degree indicating the probability/extent to which W_j applies when W applies.[7]

Definition 4 (Hyperseed reduction profile)**.** *Given a base vocabulary $B \subseteq \mathcal{H}$, a* Hyperseed reduction profile *of a concept W is any profile* $\mathrm{Prop}_B(W)$ *of the form*

$$\mathrm{Prop}_B(W) := \{(F_1, d_1), \ldots, (F_k, d_k)\}, \qquad F_i \in B,$$

intended to approximate $\mathrm{Prop}(W)$ *well enough that W becomes ϵ-scrutable from B (Definition 2) for suitably small ϵ.*

[5] Hyperseed-Concept (*Probability*) in [8].

[6] Cf. the Hyperseed concept *Contexts and Aspects* in [8].

[7] This "properties-with-degrees" form is intended as an abstract interface. In PLN, the e_j may be realized as truth values, strength/confidence pairs, or other uncertain annotations.

2.4 Theorems linking reduction quality to information and complexity

To use reductions as an *inference-control heuristic*, we need a scalar notion of how much a candidate base concept F actually helps us understand W. Information theory provides exactly this: the mutual information $I(F;W)$ is the expected reduction in uncertainty about W gained by learning F.[8]

Theorem 1 (Mutual information equals expected uncertainty reduction)**.** *Let F and W be Bernoulli random variables representing the propositions "x is F" and "x is W" under some case distribution on Ω. Let $H(\cdot)$ denote Shannon entropy. Then*

$$I(F;W) \;=\; H(W) - H(W \mid F),$$

so $I(F;W)$ is exactly the expected reduction (in bits) in uncertainty about W obtained by knowing F.

Proof. This is immediate from the definition of mutual information:

$$I(F;W) := H(W) - H(W \mid F) = H(F) + H(W) - H(F,W).$$

□

If we may pick only k Hyperseed primitives to form a reduction profile, a natural criterion is to choose those that jointly maximize information about W, thereby minimizing residual uncertainty. This connects directly to a practical PLN control strategy: route attention toward the base predicates that maximize expected information gain.

Corollary 1 (Best k-primitive reduction as an information objective)**.** *Let $S = \{F_1, \ldots, F_k\} \subseteq B$ be a set of base predicates. Then*

$$I(W;S) = H(W) - H(W \mid F_1, \ldots, F_k).$$

In particular, minimizing the residual uncertainty $H(W \mid F_1, \ldots, F_k)$ is equivalent to maximizing $I(W;S)$.

Proof. Apply Theorem 1 with F replaced by the joint variable $(F_1, \ldots, F_k)$. □

Chalmers emphasizes that a compact base must avoid *trivializing mechanisms* (e.g. encoding all truths into a single number). In an AI reduction framework, a direct way to prevent this is to penalize *description complexity* of primitives. Algorithmic information theory makes the point precise: a primitive that "encodes everything" must itself be very complex and is therefore disfavored by a minimum-description-length objective.

Theorem 2 (Complexity-regularized reduction discourages trivial encodings)**.** *Let $K(\cdot)$ denote prefix Kolmogorov complexity, and let $K(W \mid F)$ denote conditional complexity. Define the MDL-style score*

$$L(F \to W) := K(F) + K(W \mid F).$$

Then for any F,

$$L(F \to W) \geq K(W) - O(1).$$

Consequently, any "trivializing" primitive F_{triv} that makes $K(W \mid F_{\text{triv}}) = O(1)$ must satisfy $K(F_{\text{triv}}) \geq K(W) - O(1)$, hence cannot reduce the total description length below the inherent complexity of W.

[8] Hyperseed-Concept (*Mutual Information*) in [8].

Proof. A standard coding inequality for Kolmogorov complexity gives

$$K(W) \leq K(F) + K(W \mid F) + O(1),$$

because a shortest description of F together with a shortest conditional program for W given F yields a description of W. Rearranging yields the claimed lower bound on $L(F \to W)$. The final claim follows by substituting F_{triv} and using $K(W \mid F_{\text{triv}}) = O(1)$. □

Classical ontologies (including many linked-ontology systems) often use extensional "is-a" (subset) relationships. For a unified reduction framework, it is valuable to recover this as a special case of the richer intensional machinery, ensuring that linking Hyperseed to existing ontologies does not require different conceptual equipment.

Theorem 3 (Extensional inheritance as a special case of intensional inheritance)**.** *Assume that the properties in an intensional profile correspond to singleton elements and degrees are in* $\{0, 1\}$*, so that each concept corresponds to a set of instances. Then intensional inheritance reduces to extensional inheritance (a probabilistic subset/overlap relationship): knowing* $x \in F$ *informs* $x \in W$ *exactly via set overlap, and the information-based inheritance score is determined by these overlaps.*

Proof. Under the stated assumptions, each concept corresponds to a subset of instances; let $|F| = n$, $|W| = m$, and $|F \cap W| = k$ under a uniform prior on instances. Then

$$\mathbb{P}(W \mid F) = \frac{|F \cap W|}{|F|} = \frac{k}{n},$$

which is precisely the extensional overlap/subset signal. The mutual-information formulation then collapses to this set-based dependence because the only dependencies between F and W are via instance overlap. □

2.5 How these results fulfill a close variant of Chalmers' desiderata

We now summarize, point by point, how PLN-based approximate reduction to Hyperseed realizes a workable cousin of the Chalmers program:

1. **A compact base vocabulary.** Hyperseed's ontology is explicitly positioned as core scaffolding for expressing other concepts, organized into a semantic stack of concept families. Taking $B \subseteq \mathcal{H}$ (Definition 1) corresponds directly to choosing a compact set of expression families.

2. **Intensional, not merely extensional, reduction.** Reduction profiles (Definitions 3–4) treat a concept by its structured meaning-properties (intension) rather than only by its extension. This aligns with the Hyperseed commitments to *Representation* and *Mind-World Correspondence*.[9]

3. **Approximation and refinement rather than brittle definitions.** Approximate scrutability (Definition 2) is designed so that successive refinements of a reduction profile can strictly decrease ϵ by increasing the mutual information captured (Corollary 1).

4. **Avoiding trivializing mechanisms.** Theorem 2 makes precise why "encode everything into one primitive" is disfavored by a complexity-regularized reduction objective.

[9]Hyperseed concepts *Representation* and *Mind-World Correspondence* in [8].

5. **Compatibility with existing ontologies.** Theorem 3 guarantees that extensional inheritance is recovered as a special case, providing a principled route to link Hyperseed to external ontologies while still using an intensional, information-theoretic reduction score to guide inference.

6. **Direct inference-control heuristic.** Given a query about W, prioritize PLN inference steps that condition on base predicates $F \in B$ with highest estimated $I(F; W)$, since this yields maximal expected uncertainty reduction (Theorem 1) per inference step.

Bottom line. Chalmers' project can be read as claiming that, in an idealized epistemic setting, meanings and truths can be systematically recovered from a compact base. Hyperseed provides a candidate ontology-level base of concept families; PLN provides uncertainty-aware inference; and intensional inheritance provides a mathematically grounded metric for constructing and refining reductions. Together these deliver a detailed, workable realization of a close variant of the Chalmers desiderata.

3 Using Hyperseed to Guide PLN Inference Control

PLN inference is powerful but combinatorially explosive: at any moment there are many possible rule instances to apply and many possible premises to combine, and naive search rapidly wastes resources. Hyperseed offers a way to turn the knowledge base into a *control surface*: it provides a structured notion of semantic proximity, abstraction level, and cross-domain resonance that can be used to estimate which candidate inference actions are likely to yield progress.

From a Hyperseed perspective, inference control is a special case of attention allocation: the system must allocate limited genenergy across many possible operations.[10] The most direct quantitative idea is:

> Prefer inference steps with high *expected marginal information gain* (about what we care about) per unit *effort cost*.

The GTGI-inspired machinery makes this analogy crisp: we can treat an inference episode as a context with an associated resource-consumption distribution, and interpret "good inference control" as maximizing a competence-per-resource functional in the spirit of efficient pragmatic general intelligence, specialized to the "environment" constituted by the current knowledge base, rule set, and query.[11]

Practically, the heuristics split into two partly separable problems:

- **Rule selection:** which inference rule schema (and high-level instantiation pattern) to try next—often best learned from prior inference trajectories.
- **Premise selection:** given a rule schema, which candidate premises to bind into it—often best guided by relevance signals and association-spreading dynamics (attention as resource allocation).

Below we present three control templates—backward chaining, forward chaining, and a bidirectional geodesic methodology—each designed so that Hyperseed can plug in as a guide to expected marginal information gain.

[10] Hyperseed-Concept 60 (*Attention*) and Hyperseed-Concept 100 (*Genenergy / Effort*).

[11] Hyperseed-Concept 214 (*Context*), Hyperseed-Concept 216 (*Resource-Consumption Distribution*), and Hyperseed-Concept 217 (*Efficient Pragmatic General Intelligence*).

3.1 Hyperseed-guided backward chaining

Backward chaining begins from a target conclusion and works upstream toward premises. In PLN this means: given a query Q, select a rule that could derive (or strengthen) Q, then select subgoals whose satisfaction would enable that rule instance, and iterate.

Hyperseed enters in two places: (i) it supplies semantic decompositions of Q into nearby/related concepts, suggesting plausible intermediate subgoals; and (ii) it helps estimate which subgoals are likely to be *informative* about Q relative to the current belief state, i.e., which ones reduce uncertainty in distinctions relevant to Q.[12]

Technical sketch. Let B denote the current belief state (a set of PLN truth-valued expressions), and let Q denote the current query. Consider a candidate backward step a that replaces Q by a multiset of subgoals $\{G_1, \ldots, G_m\}$ together with a planned rule instance schema σ such that $\sigma(G_1, \ldots, G_m) \Rightarrow Q$.

Define a scoring functional

$$\mathrm{Score}_{\leftarrow}(a; B, Q) \ := \ \frac{\mathbb{E}[\Delta \mathcal{I}_{\mathrm{HS}}(B, Q \mid a)]}{\mathrm{Cost}(a)}, \tag{1}$$

where:

- $\mathcal{I}_{\mathrm{HS}}$ is any Hyperseed-indexed information/relevance functional—for example, entropy reduction over a Hyperseed-induced partition of conceptual space, or a Hyperseed graph-distance-weighted estimate of uncertainty reduction. This reflects the "observer-indexed bookkeeping" stance used throughout Hyperseed: distinctions drive what counts as information, and beliefs weight those distinctions.[13]
- $\mathrm{Cost}(a)$ is an effort/genenergy estimate[14]—e.g. CPU time, memory bandwidth, or proof-search expansion budget.
- The expectation is taken over uncertain bindings and uncertain premise truth-values (PLN-style).

The control policy is: choose a maximizing $\mathrm{Score}_{\leftarrow}(a; B, Q)$, subject to budget constraints.

Implementation notes. In practice the pieces in (1) are best estimated by different mechanisms at different granularities:

- **Rule selection as probabilistic program generation.** Maintain a learned prior over rule schemas and macro-patterns of chaining, drawn from saved inference histories. Hyperseed can index and condition this prior by ontological type, abstraction level, and analogy class.[15]
- **Premise selection via association-spreading (ECAN-like).** Given a rule schema, select premises by spreading activation from the current query and/or subgoals through the knowledge graph. Hyperseed provides the high-level semantic scaffolding for what edges exist and how activation should decay across them. This is attention allocation in the literal Hyperseed sense, bounded by resource budgets.[16]

[12] Subgoal informativeness is assessed via Hyperseed-Concept 139 (*Probabilistic Belief*) and Hyperseed-Concept 98 (*Information as Distinction*).

[13] Hyperseed-Concept 98 (*Information as Distinction*) and Hyperseed-Concept 139 (*Probabilistic Belief*).

[14] Hyperseed-Concept 100 (*Genenergy / Effort*).

[15] Cognitive synergy heuristics align here with Hyperseed-Concept 73 (*Cognitive Synergy*).

[16] Hyperseed-Concept 60 (*Attention*) and Hyperseed-Concept 100 (*Genenergy / Effort*).

3.2 Hyperseed-guided forward chaining

Forward chaining begins from what is currently believed and pushes consequences outward: pick a rule and a set of supported premises, derive a conclusion, and continue. It is essential for continual inference in embodied or domain-focused systems, since it can opportunistically derive useful results before a particular query arrives and can update the belief state as new percepts arrive.

Hyperseed helps forward chaining avoid degenerating into pointless closure computation by biasing inference toward concept regions and abstraction levels likely to be useful for the system's active goals. In GTGI terms, the environment and goals define a context, and forward inference tries to increase expected goal-achievement per unit resource.[17]

Technical sketch. Let B be the current belief state and let G be a set of currently active goals/queries (possibly empty). A candidate forward step a is a rule instance $\sigma(P_1, \ldots, P_k) \Rightarrow C$ applied to premises $P_i \in B$, yielding a new conclusion C added to B with an updated truth value.

Define

$$\mathrm{Score}_{\rightarrow}(a; B, G) \;:=\; \frac{\mathbb{E}[\Delta\mathcal{U}_{\mathrm{HS}}(B, G \mid a)]}{\mathrm{Cost}(a)}, \tag{2}$$

where $\mathcal{U}_{\mathrm{HS}}$ is a Hyperseed-indexed utility functional measuring expected future usefulness of adding C to the belief state, relative to active goals G and/or expected goal distributions (observer-indexed distinctions weighted by probabilistic belief).[18]

Common specializations include:

- *Goal-directed:* $\mathcal{U}_{\mathrm{HS}}$ measures expected increase in support for members of G (or decrease in their uncertainty).
- *Curiosity / coverage:* $\mathcal{U}_{\mathrm{HS}}$ measures expected increase in conceptual breadth/coverage in a Hyperseed-induced partition of conceptual space—naturally resource-sensitive via genenergy budgeting.[19]

Embodiment and domain anchoring. For an embodied AI system (or any system specialized to a domain), Hyperseed-guided forward inference depends critically on *background inference links* connecting Hyperseed concepts to sensorimotor primitives, task affordances, domain ontologies, instrumentation vocabularies, and so on. These links can be hand-built, learned, or imported (e.g. via mappings from external ontologies); once present, they function as bridges allowing PLN to translate abstract Hyperseed-level inferences into concrete, actionable domain conclusions.

A simple but important monotonicity observation is:

Proposition 1 (Adding sound Hyperseed–domain bridges cannot reduce attainable inference performance)**.** *Fix a PLN rule set and a resource budget. Suppose we extend a knowledge base B to $B' := B \cup \Delta$ by adding a set Δ of* sound *bridge axioms/rules connecting Hyperseed concepts to domain or embodiment concepts. If the inference controller is allowed to ignore any added axiom/rule and the evaluation of goal-achievement is monotone in the available set of admissible proof paths, then the maximum achievable expected goal-achievement using B' is at least that using B (for the same resource budget).*

[17] Hyperseed-Concept 214 (*Context*) and Hyperseed-Concept 217 (*Efficient Pragmatic General Intelligence*).
[18] Hyperseed-Concept 98 (*Information as Distinction*) and Hyperseed-Concept 139 (*Probabilistic Belief*).
[19] Hyperseed-Concept 100 (*Genenergy / Effort*).

Proof sketch. Any inference trajectory available in B is still available in B' (simply never use the added bridges). By the assumed monotonicity of evaluation with respect to the available proof-path set, the optimum over the larger admissible set cannot be smaller than the optimum over the subset. □

The point is not that bridges automatically help; it is that, under sane control assumptions, adding *correct* bridges expands the reachable conceptual territory without forcing degradation. The real engineering work is making the controller prefer bridges when they offer good expected marginal gain per unit cost.[20]

3.3 Bidirectional geodesic search

Forward and backward chaining are asymmetric: one pushes from what is known, the other pulls from what is wanted. In complex inference problems, either extreme can thrash: forward chaining can diffuse effort into irrelevant consequences, and backward chaining can generate subgoals that are hard to ground in available premises. A natural alternative is to work both ends at once, attempting to connect them in the middle.

Geodesic inference control provides a principled picture of this bidirectional strategy: view inference as searching for a least-effort path between an *initial* distribution (premises/priors) and a *goal* distribution (desired conclusions/queries), with effort measured in a resource-sensitive way. The Schrödinger-bridge view yields a canonical form of intermediate guidance: weight intermediate states by a product of a forward factor (reachability from the past) and a backward factor (usefulness toward the future). This directly motivates bidirectional chaining: grow frontiers from both ends and prefer steps that improve the product per unit effort.

Technical sketch. Let X be a space of candidate intermediate inference states (e.g. partial proofs, partial bindings, or subgoal sets). Let ρ_{init} be an initial distribution over X induced by the current premises, and let ρ_{goal} encode what "being close to the goal" means (e.g. states that already entail the query with high support). A geodesic controller maintains two frontiers:

- a forward frontier grown from ρ_{init} (premise-side reachability),
- a backward frontier grown from ρ_{goal} (goal-side pull).

Introduce two potentials $\phi := \log f$ and $\psi := \log g$, where heuristically:

- $f(x,t)$ measures how easily the forward process can reach x by time t,
- $g(x,t)$ measures how easily x can continue to the goal by time t.

A generic step-selection score is then

$$\text{Score}_{\text{geo}}(\text{step } s) \;:=\; \Delta(\phi+\psi) \;-\; \lambda\,\Delta\text{Cost}(s), \tag{3}$$

where $\lambda > 0$ trades off benefit against effort cost.

A minimal pseudocode sketch:

[20] I.e., the controller must treat bridge-use as an attention-allocation decision—Hyperseed-Concept 60 (*Attention*)—subject to genenergy budgets—Hyperseed-Concept 100 (*Genenergy / Effort*).

```
init forward frontier F with rho_init
init backward frontier B with rho_goal
while budget not exhausted:
  choose a side to expand (to keep step-costs roughly equal)
  for each candidate step s from chosen frontier node x:
    estimate phi(x_next), psi(x_next)    # from f,g surrogates
    score[s] = (phi+psi at x_next) - (phi+psi at x) - lambda*cost(s)
  take top-K steps, update frontier estimates
  attempt to splice a forward and backward partial path that meet/unify
return best spliced paths found
```

Approximating the forward and backward factors. In practice f and g are estimated by statistical and structural surrogates:

- local message passing when the inference constraints factorize (factor-graph style),
- short-horizon Monte Carlo "meet" probabilities between frontiers,
- amortized predictors trained from prior inference runs (learned ϕ, ψ).

Hyperseed enters here by (i) defining which semantic factorization is appropriate (ontology-induced structure), and (ii) shaping the features used by learned predictors (ontological type, abstraction depth, archetype adjacency, etc.).

Why the product guidance is principled. A key conceptual reason to prefer bidirectional geodesic guidance over ad hoc mixtures is that the product $f \cdot g$ is the canonical time-symmetric way to weight intermediate states between boundaries:

Proposition 2 (Time-symmetric intermediate weighting has product form)**.** *In a Schrödinger-bridge (least-effort) interpolation between boundary distributions, the time-t intermediate weighting that remains consistent under arbitrary subdivision of the interval is (up to normalization) the product of a forward factor $f(\cdot, t)$ and a backward factor $g(\cdot, t)$:*

$$\rho(x, t) \propto f(x, t)\, g(x, t).$$

Proof sketch. Schrödinger-bridge theory yields a unique bridge marginal at intermediate time t determined by the boundary data and the reference dynamics. This bridge marginal factorizes into a forward and backward solution of the corresponding scaling equations; the factorization implies time symmetry, and the uniqueness implies interval consistency. The product form is therefore not an arbitrary heuristic but the canonical intermediate weighting induced by the boundary-value, least-effort formulation. □

From the Hyperseed perspective, the significance is that the controller implements a resource-sensitive attention policy that is *structurally* compatible with two-ended guidance: it avoids both pure diffusion (forward-only) and pure fantasy (backward-only), and it naturally fits the cognitive-synergy theme because different subsystems can specialize in estimating different parts of the guidance signal.[21]

[21] Hyperseed-Concept 73 (*Cognitive Synergy*), Hyperseed-Concept 60 (*Attention*), and Hyperseed-Concept 100 (*Genenergy / Effort*).

4 Structural Conditions for Exponential Speedup in Ontology-Guided Inference

The preceding sections established that Hyperseed can serve as a scrutability-style base (Section 2) and that its structure can guide PLN inference control (Section 3). The theorems presented so far, however, are essentially foundational identities and monotonicity arguments: they show that the framework is *coherent* and that adding ontological structure *cannot hurt*, but they do not explain when it provides a *dramatic* advantage. A natural question remains: under what conditions does ontology-guided inference achieve exponentially better performance than unguided search, and how can one empirically verify that those conditions hold?

This section answers both questions. We introduce stochastic structural conditions on the relationship between the Hyperseed base and the broader concept space, prove that these conditions imply exponential search-cost reduction, characterize the conditions graph-theoretically via an approximate separator / ontological treewidth framework, and describe a diagnostic program for empirically measuring whether the conditions hold in a deployed system. We then illustrate the theory with two worked examples.

4.1 Inference as tree search: the two-level picture

PLN inference can be modeled as exploring a search tree $\mathcal{T}$ of depth D and per-node branching factor up to b. At each internal node v (representing a partial inference state), the system chooses among $\mathrm{Ch}(v)$ candidate next steps (rule instantiations and premise bindings). The value function $J^*(v) = \max_{w \in \mathrm{Ch}(v)}[g(v,w) + J^*(w)]$ satisfies the Bellman equation, where $g(v,w)$ is the net gain of a single inference step. Exhaustive search costs $O(b^D)$.

Each inference state v admits a *Hyperseed projection* $\pi_B(v) \in \mathbb{R}^n$ (its reduction profile with respect to the ontological base B, $|B| = n \ll N$). The central question is whether $\pi_B(v)$ is informative enough to guide search.

The realistic picture is a *two-phase* decision at each node:

1. **Ontological triage (cheap).** Use π_B to partition $\mathrm{Ch}(v)$ into *ontological clusters*—groups of children that look similar from the abstract-ontological standpoint—rank the clusters by projected value, and discard clusters that are clearly unpromising.

2. **Concrete discrimination (moderate cost).** Within each surviving cluster, use domain-specific data and local reasoning to identify the best child.

The total cost per level is (number of surviving clusters) $\times$ (cost to resolve each cluster), iterated over D depth levels.

This two-level structure reflects an important asymmetry. An abstract ontology like Hyperseed can distinguish *types* of inference actions (e.g. "morphological comparison" vs. "chemical analysis" for an animal-classification query) but cannot distinguish specific instances within a type (e.g. "compare ear length" vs. "compare tail shape"). The former is ontological triage; the latter is concrete discrimination requiring domain data.

4.2 Stochastic structural conditions

In any realistic deployment the system faces a *distribution* $\mathcal{D}$ of inference problems. We state conditions under the *node-visit distribution* μ: the distribution over nodes actually encountered by the guided controller during inference on $\omega \sim \mathcal{D}$. This is important—conditions need not hold at nodes the controller never visits.

At each visited node v, the Hyperseed clustering yields $K(v)$ clusters, $q(v)$ of which survive triage, with per-cluster concrete-resolution costs $c_k(v)$.

Definition 5 (Stochastic Ontological Cluster Separation, SOCS$(\bar{\beta}, \tau)$)**.** *For each cluster $C_k(v)$ let $V_k^{\max}(v) = \max_{w \in C_k}[g(v, w) + J^*(w)]$ and let $\hat{V}_k(v)$ be its Hyperseed-projected value estimate. Define the* local OCS failure event*:*

$$\mathrm{Fail}(v) \;=\; \{\exists\, j, k : \; \hat{V}_j > \hat{V}_k + 3\bar{\beta} \;\;\textit{but}\;\; V_j^{\max} \le V_k^{\max}\}.$$

We require:

(a) $\Pr_{v \sim \mu}[\mathrm{Fail}(v)] \le \tau$ *(low failure rate).*

(b) $\mathbb{E}_{v \sim \mu}[\beta(v)] \le \bar{\beta}$ *(controlled average tolerance), where $\beta(v)$ is the smallest β for which OCS holds at v.*

This condition says: the ontological projection usually ranks clusters correctly, but is allowed to be confused at a τ-fraction of nodes and to have tolerance up to $\bar{\beta}$ on average. It does *not* require the projection to rank children *within* a cluster—that is what concrete reasoning handles.

Definition 6 (Stochastic Cluster Tractability, SCT$(\bar{q}, \bar{c}, \sigma_q, \sigma_c)$)**.** *Under μ:*

$$\mathbb{E}[q(v)] \le \bar{q}, \quad \mathrm{Var}[q(v)] \le \sigma_q^2, \quad \mathbb{E}\Big[\textstyle\sum_{k\,\mathrm{retained}} c_k(v)\Big] \le \bar{c}, \quad \mathrm{Var}\Big[\textstyle\sum_{k\,\mathrm{retained}} c_k(v)\Big] \le \sigma_c^2.$$

Note that cluster sizes do not appear as a separate condition; they are folded into $c_k(v)$. A large cluster that is easy to resolve (concrete features are very discriminating) is fine. A small cluster that is expensive to resolve is also fine if it is atypical. What matters is the *total cost distribution.*

Definition 7 (Projection Efficiency, PE$(p(n))$)**.** *Computing $\hat{V}_k$ for all $K(v)$ clusters at node v costs at most $K(v) \cdot p(n)$ operations, where p is polynomial in the Hyperseed base size n.*

4.3 Main theorem: stochastic two-level ontological inference efficiency

Theorem 4 (Stochastic Two-Level Ontological Inference Efficiency)**.** *Let $\mathcal{D}$ be a distribution over inference problems with trees of depth at most D and per-node branching factor at most b. Assume SOCS$(\bar{\beta}, \tau)$, SCT$(\bar{q}, \bar{c}, \sigma_q, \sigma_c)$, and PE$(p(n))$. Consider the two-level guided controller $\mathcal{A}_{H+C}$ that at each node performs ontological triage (retaining clusters within $3\bar{\beta}$ of the top projected value) followed by concrete discrimination within each retained cluster.*

(a) Expected regret (greedy controller). *If the controller commits greedily to the best-estimated child at each level,*

$$\mathbb{E}_{\omega \sim \mathcal{D}}\big[J^*(s_0) - J^{\mathcal{A}}(s_0)\big] \;\le\; D\big(6\bar{\beta} + R_{\max}\,\tau\big)$$

where $R_{\max}$ bounds single-step gain.

(b) High-probability regret. *With probability $\ge 1 - \delta$:*

$$J^*(s_0) - J^{\mathcal{A}}(s_0) \;\le\; D\Big(6\bar{\beta} + R_{\max}\tau\Big) + O\Big(R_{\max}\sqrt{D \ln(1/\delta)}\Big).$$

(c) Expected computational cost.

$$\mathbb{E}[\mathrm{Cost}(\mathcal{A}_{H+C})] \;\le\; D\big(\bar{K}\, p(n) + \bar{c}\big)$$

where $\bar{K} = \mathbb{E}[K(v)]$.

(d) Exponential speedup. *The expected speedup over exhaustive search is*

$$\frac{b^D}{D(\bar{K}\,p(n) + \bar{c})},$$

which is exponential in D whenever the per-node guided cost $\bar{K}\,p(n) + \bar{c}$ is bounded.

(e) Exact optimality (multi-path variant). *If the controller recurses on* all *$q(v)$ surviving cluster-winners at each level, it finds the exactly optimal path with probability $\geq 1 - \tau D$. The expected number of nodes explored is at most q_{eff}^D, where $q_{\mathrm{eff}} = \bar{q} + \sigma_q^2/\bar{q}$ accounts for variance across levels. The expected speedup is $(b/q_{\mathrm{eff}})^D$.*

Proof. **Part (a).** At each level ℓ, let w_ℓ^* be the truly optimal child and $\tilde{w}_\ell$ the controller's selection. Single-level regret R_ℓ splits into two cases.

Case 1: no OCS failure (probability $\geq 1 - \tau$). The cluster containing w_ℓ^* is retained: if it were excluded, some other cluster would be confidently ranked above it, yielding $\hat{V}_j > \hat{V}_{k^*} + 3\bar{\beta}$ with $V_j^{\max} \leq V_{k^*}^{\max}$, contradicting SOCS. All retained cluster-winners have projected values within $3\bar{\beta}$ of each other (by the retention threshold), so their true max-values differ by at most $6\bar{\beta}$ (accounting for $\bar{\beta}$ tolerance in both directions). Thus $R_\ell \leq 6\beta(v_\ell)$, and $\mathbb{E}[R_\ell \cdot \mathbf{1}[\text{no fail}]] \leq 6\bar{\beta}$.

Case 2: OCS failure (probability $\leq \tau$). $R_\ell \leq R_{\max}$, so $\mathbb{E}[R_\ell \cdot \mathbf{1}[\text{fail}]] \leq R_{\max}\tau$.

Summing: $\sum_\ell \mathbb{E}[R_\ell] \leq D(6\bar{\beta} + R_{\max}\tau)$.

Part (b). The failure indicators $\mathbf{1}[\text{fail at } v_\ell]$ are Bernoulli with parameter $\leq \tau$. Their sum F satisfies $\Pr[F > D\tau + \sqrt{D \ln(1/\delta)/2}] \leq \delta$ by Hoeffding's inequality. Similarly, $\sum_\ell \beta(v_\ell)$ concentrates around $D\bar{\beta}$. A union bound gives the stated high-probability bound.

Part (c). Per level the controller computes cluster projections $(K(v) \cdot p(n))$ and resolves retained clusters $(\sum_{k\,\text{ret.}} c_k(v))$. Expectations under μ give the bound.

Part (d). Ratio of b^D to the Part (c) bound.

Part (e). At each level the optimal child is among the $q(v)$ survivors (when OCS does not fail); the failure probability over D levels is $\leq \tau D$ by a union bound. For the node count, if $q(v_\ell)$ are independent across levels then $\mathbb{E}[\prod_\ell q(v_\ell)] = \bar{q}^D$; under positive correlations the bound inflates to q_{eff}^D with $q_{\mathrm{eff}} = \bar{q} + \sigma_q^2/\bar{q}$ (via the identity $\mathbb{E}[q^2] = \bar{q}^2 + \sigma_q^2$ and a product-moment inequality). $\square$

Remark 1. *The* Ontological Efficiency Ratio $\mathrm{OER} = \log \bar{q} / \log b$ *is the key summary statistic. The exponential speedup factor is $b^{(1-\mathrm{OER})\cdot D}$.* $\mathrm{OER} = 0$ *means perfect discrimination (only one cluster survives);* $\mathrm{OER} = 1$ *means no discrimination at all.*

4.4 Graph-theoretic characterization: ontological separation and treewidth

The preceding theorem assumes SOCS. A natural follow-up question is: *what structural property of the concept graph causes SOCS to hold?* The answer is that the Hyperseed base acts as an approximate separator—a "thin waist" through which most long-range conceptual dependencies flow.

Definition 8 (Concept relevance graph)**.** *Let $\mathcal{C} = \{C_1, \ldots, C_N\}$ be the working concept space. The* concept relevance graph *$G_\eta = (\mathcal{C}, E, w)$ has edge weights $w(i,j) = I(C_i; C_j)$ (mutual information under the system's belief distribution), thresholded at $\eta > 0$.*

Definition 9 (Approximate Ontological Separation, AOS(ϵ, s))**.** *B is an (ϵ, s)-approximate separator if, upon removing B from G_η:*

1. *the residual graph $G_\eta[\mathcal{C} \setminus B]$ decomposes into connected components $\mathcal{R}_1, \ldots, \mathcal{R}_L$ of size at most s;*
2. *the* leakage*—total weight of edges between distinct residual components—satisfies* $\mathrm{Leak}(B) := \sum_{i \in \mathcal{R}_a,\, j \in \mathcal{R}_b,\, a \neq b} w(i,j) \leq \epsilon\, W_{\mathrm{total}}$*, where* $W_{\mathrm{total}} = \sum_{(i,j) \in E} w(i,j)$*.*

Condition 1 says that without the ontological scaffold the concept space fragments into small domain-specific islands. Condition 2 says that cross-island information flow bypassing the ontology is a small fraction ϵ of the total.

The separator property implies an approximate conditional-independence structure:

Proposition 3 (Separation implies approximate Markov blanket)**.** *If B satisfies AOS(ϵ, s) and $\mathbb{P}$ satisfies the local Markov property with respect to G_η, then for all $a \neq b$,*

$$I(\mathcal{R}_a; \mathcal{R}_b \mid B) \;\leq\; O\!\left(\frac{\mathrm{Leak}(B)}{\eta}\right) \;=\; O(\epsilon\, W_{\mathrm{total}}/\eta).$$

Proof sketch. Under the local Markov property, exact graph separation implies exact conditional independence. Leakage edges contribute residual conditional dependence bounded, via the data processing inequality applied per edge, by the total leakage weight divided by the threshold η. □

This approximate Markov blanket property is what makes SOCS hold: the Hyperseed projection captures most of the inter-regional dependence structure, so it can correctly rank clusters of inference actions that exploit different concept regions.

Definition 10 (Ontological treewidth)**.** *$\mathrm{tw}_B(G_\eta)$ is the treewidth of the residual graph $G_\eta[\mathcal{C} \setminus B]$.*

Theorem 5 (Separation implies SOCS; treewidth controls concrete cost)**.** *Let G_η be the concept relevance graph with Hyperseed base B satisfying AOS(ϵ, s), and let $t = \mathrm{tw}_B(G_\eta)$. Suppose $\mathbb{P}$ satisfies the local Markov property and the inference value function decomposes additively over concept-footprint contributions:*

$$g(v, w) + J^*(w) \;=\; \phi_B(w) + \sum_{a : \mathcal{R}_a \cap F(w) \neq \emptyset} \phi_a(w) + \xi(w)$$

where ϕ_B depends only on $\pi_B(w)$, each ϕ_a depends on local state within $\mathcal{R}_a$, and ξ captures cross-component leakage. Then:

(i) $\mathbb{E}[|\xi|^2]^{1/2} \leq O(\sqrt{\epsilon}\, \|\nabla_{\mathcal{C}} J^*\|)$*, where $\|\nabla_{\mathcal{C}} J^*\|$ is a Lipschitz constant of the value function.*

(ii) *SOCS$(\bar{\beta}, \tau)$ holds with*

$$\bar{\beta} \leq O\Big(\sqrt{\epsilon}\, \|\nabla_{\mathcal{C}} J^*\| + \mathbb{E}_\mu[\max_a \mathrm{range}(\phi_a)]\Big), \qquad \tau \leq O\!\left(\frac{\epsilon\, \|\nabla_{\mathcal{C}} J^*\|^2}{(\text{typical inter-cluster gap})^2}\right).$$

(iii) *Concrete-resolution cost within a residual component $\mathcal{R}_a$ satisfies $c(\mathcal{R}_a) \leq O(|\mathcal{R}_a| \cdot \exp(t))$. When $t = O(\log N)$ (logarithmic ontological treewidth), this cost is polynomial in N.*

Proof sketch. **(i)** For an inference step with footprint spanning components $\mathcal{R}_a, \mathcal{R}_b$, the leakage term ξ arises because $\mathbb{P}(\mathcal{R}_a, \mathcal{R}_b \mid B)$ is not exactly factored. By Proposition 3 and Pinsker's inequality the total-variation distance is $O(\sqrt{\epsilon})$; the value perturbation is bounded by $\|\nabla_{\mathcal{C}} J^*\|$ times this distance.

(ii) The projected value $\hat{V}_k$ estimates $\phi_B^{(k)}$. From the additive decomposition, inter-cluster value discrepancy equals the ontological gap $(\phi_B^{(j)} - \phi_B^{(k)})$ plus concrete and leakage residuals bounded by $\max_a \mathrm{range}(\phi_a) + \mathrm{range}(\xi)$. The leakage contribution to $\bar{\beta}$ is $O(\sqrt{\epsilon}\,\|\nabla J^*\|)$ from part (i). For τ: ranking failures occur when residuals exceed the inter-cluster ontological gap; by Chebyshev the failure probability is $O(\epsilon\|\nabla J^*\|^2/\mathrm{gap}^2)$.

(iii) Standard results in graphical-model inference: exact inference in a model with treewidth t costs $O(n_{\mathrm{vars}} \cdot d^{t+1})$, where d is the per-variable domain size. □

Interpretation. Theorem 5 explains *why* abstract ontological concepts are natural graph separators. Abstraction is precisely the cognitive operation of identifying structure shared across domains. Shared structure manifests as hub nodes in the concept graph: "morphological pattern," "causal process," "distinction," "representation" mediate dependencies between otherwise disparate domain-specific concept clusters. Remove these hubs and the graph fragments into domain islands.

4.5 Diagnostic program

All conditions are empirically testable from inference traces and the knowledge base.

Ontological Efficiency Ratio (OER). $\widehat{\mathrm{OER}} = \log \hat{\bar{q}} / \log \bar{b}$. This is the headline number: the expected speedup is approximately $\bar{b}^{(1-\widehat{\mathrm{OER}})\cdot D}$.

OCS violation rate $\hat{\tau}$. At each visited node, check whether the ontological cluster ranking agrees with the true value ranking for cluster pairs whose gap exceeds $3\hat{\bar{\beta}}$. If $\hat{\tau} > 0.05$, the ontology has systematic blind spots.

Distribution of surviving clusters $q(v)$. Heavy tails indicate specific inference contexts where the ontology provides little discrimination. Identifying these contexts indicates where ontological enrichment would be most valuable.

Regret decomposition. Decompose observed regret into the two sources: (i) imprecision (non-optimal selection among retained clusters, bounded by $6\bar{\beta}$ per level) and (ii) failure (OCS violations, bounded by $R_{\max}\tau$ per level). Imprecision is benign and expected; failure indicates structural problems.

Residual component analysis. Remove B from G_η and compute connected component sizes. Large components are candidates for ontological enrichment. For each component, compute the ratio of internal to boundary edges; low ratios indicate poor ontological mediation.

Marginal value of ontological expansion. Progressively add concepts to B and re-estimate OER. Plot $\widehat{\mathrm{OER}}(n)$ as a function of $|B| = n$. The plateau point indicates the "natural complexity" of the ontological base: beyond it, remaining variation is inherently concrete.

Cross-level correlation. Measure $\mathrm{Corr}(q(v_\ell), q(v_{\ell+1}))$ across successive levels. High correlation means some problems are systematically harder (the ontology is less useful for them); the Version A search cost inflates above $\bar{q}^D$.

Spectral coverage (a priori check). Compute the mutual-information matrix $\mathbf{M}$ and measure how much spectral mass the B-columns capture: $\mathrm{SC}(B) = \sum_{i=1}^{n} \sigma_i(\mathbf{M}_B)^2 / \sum_{j=1}^{N} \sigma_j(\mathbf{M})^2$. High spectral coverage is necessary (but not sufficient) for small $\bar{\beta}$.

5 Illustrative Examples

5.1 Illustrative example I: integrative educational diagnosis

To see how ontology-driven reasoning might work in practice, consider an example everyday scenario: an AI tutoring system helping a struggling 15-year-old student. The surface presentation is failing math/science grades, incomplete homework, increasing absenteeism. The reasoning task is to diagnose what is actually going wrong and construct a coherent intervention.

The domains involved. This task spans six concept clusters: (i) academic/cognitive (prerequisite chains, working memory, reading comprehension), (ii) motivational /psychological (self-efficacy, anxiety-avoidance cycles, learned helplessness), (iii) social/relational (peer dynamics, teacher relationships, family pressure), (iv) temporal/developmental (adolescent identity formation, how current choices affect future options), (v) resource/practical (limited study time, competing demands), and (vi) pedagogical (scaffolding, zone of proximal development, spaced repetition).

Without ontological scaffolding, the system faces $b \approx 15$–25 plausible interventions at each decision point, each grounded in a different domain's logic. Evaluating cross-domain interactions ("Will drilling fractions help if anxiety makes her shut down during math?") requires bridging between clusters.

Hyperseed concepts as separators. The Hyperseed concept of *distinction* bridges the academic and motivational domains: the key diagnostic question—"Is the student failing because she *cannot* do the math, or because she *will not* attempt it?"—is a distinction that is graded, observer-relative, and evidence-based (probeable by offering scaffolded problems at varying difficulty and observing engagement patterns). Without this bridge the system would need to independently discover that the academic assessment question and the motivational assessment question are two aspects of a single diagnostic distinction.

Wu wei and effort bridge the resource and intervention domains: the best intervention is one where the student's own default tendencies are already partially aligned with the goal, requiring only gentle steering. Drilling fractions against active anxiety means high forcing cost; first addressing the anxiety aligns P_0 with the learning task, making subsequent instruction low-forcing.

Attention bridges the cognitive and scheduling domains: study time, emotional bandwidth, and processing capacity are all instances of resource allocation under constraints—a single Hyperseed-structured optimization rather than three separate domain-specific problems.

Representation and resonance bridge the social and intervention domains: the student's self-concept ("I'm not a math person") is a representation that is maintained by social resonance; changing it requires feedback from trusted others that aligns with and gradually shifts her self-model.

Efficiency analysis. With $b \approx 20$ and $q \approx 3$, OER ≈ 0.37—the same as the creativity example, since both are cross-domain bridging tasks. Over $D \approx 5$ major reasoning steps, the speedup is $20^{(1-0.37)\times 5} \approx 10^4$.

Where the ontology helps less. For purely intra-domain questions—e.g. "What is the best way to teach fraction addition to a 15-year-old who understands the concept but makes computational errors?"—the OER would be close to 1: Hyperseed's abstract concepts add little beyond what the pedagogical domain's internal structure provides. The theory predicts that Hyperseed-guided inference shows its largest advantages for integrative, cross-domain reasoning, exactly the type of reasoning most needed by a general intelligence.

Diagnostic implication. The residual component analysis (Subsection 4.5) would reveal which domains are well-separated by Hyperseed and which need enrichment. In this example, the social/relational domain might show higher leakage ϵ (some direct dependencies between peer influence and motivation that bypass Hyperseed), suggesting that additional ontological concepts in the social layer would be valuable.

5.2 Illustrative example II: cross-domain creative reasoning

As a different sort of illustrative example, we consider a reasoning task that we actually carried out as part of our own AI research: the construction of the *Motivated Wu-Wei Creativity Hypothesis* [9], which argues that under resource constraints and repeated sociocultural interaction, AI systems whose creative behavior is driven by internally grounded motivational dynamics will participate in creative practice more efficiently than motivation-less prompting approaches.

Why the task is hard without ontological scaffolding. The argument bridges five distinct conceptual domains normally studied separately: (i) aesthetics / philosophy of art, (ii) KL-regularized optimal control, (iii) motivational psychology (PSI theory), (iv) social/cultural theory, and (v) generative AI. Without ontological structure, formalizing the initial informal intuition requires searching across the full product space of formalisms from each domain; with branching factor $b \approx 20$ (plausible formalisms per step) and depth $D \approx 6$ (major reasoning stages), the unguided search space is $\sim 20^6 \approx 6 \times 10^7$ candidate paths.

Hyperseed concepts as separators. In the concept relevance graph, the five domain clusters are connected almost entirely through Hyperseed bridge concepts:

Wu wei / minimal-forcing control mediates between the control-theory cluster and the aesthetics cluster: the Hyperseed formalization of effort as transport cost plus KL-forcing cost directly yields the paper's core formal object $J(\pi;\alpha) = \mathbb{E}[\sum_t c_\alpha(x_t)] + \lambda\,\mathbb{E}[\sum_t \mathrm{KL}(P_\pi \| P_0)]$.

Attention and genenergy/effort mediate between motivational psychology and control theory: the insight that PSI-style modulators shape the passive dynamics P_0 (so that a well-motivated system pays low forcing cost) falls out directly from the Hyperseed framework.

Resonance mediates between cultural theory and the control/AI clusters: social feedback becomes a signal that drives modulator recalibration, adjusting P_0 and reducing future forcing cost for culturally aligned creative behavior.

Pattern, repetition, and distinction mediate between aesthetics and the formalization of intention: creative intention becomes "a temporally extended pattern of stable goal selection, coherent modulator trajectories, and feedback-sensitive revisions that preserve project identity."

Efficiency analysis. At each reasoning step, ontological triage reduces $b \approx 20$ candidates to $q \approx 2$–3 (the ontologically indicated cluster plus one or two alternatives). Over $D = 6$ steps the guided search explores $\sim 3^6 = 729$ paths vs. $\sim 6 \times 10^7$ unguided—a speedup of roughly 10^5, or OER $\approx \log 3 / \log 20 \approx 0.37$.

The concrete intra-cluster work remains: after *wu wei* identifies KL-regularized control as the right cluster, the system still needs domain expertise to choose KL divergence over Wasserstein, derive the exponential tilting theorem, etc. This is the Phase 2 concrete discrimination—bounded by the cluster size, not the full branching factor.

6 Why an Explicit Core Ontology is Valuable for AGI

Many intelligent systems develop an implicit core ontology because the world and the body impose regularities: sensors and actuators carve experience into stable dimensions, goals impose repeated evaluation pressures, and social interaction channels stabilize shared categories. In modern machine learning, this core ontology is often implicit in a learned representation: present, but not directly available as an object of deliberation.

Hyperseed makes a different bet: it is worthwhile to represent a core concept network explicitly in a manipulable, logical/relational form and then use it as a scaffold for uncertain inference and for self-modeling. The AGI-relevant claim is not that explicit ontologies are magically "better" than learned representations; it is that *explicitness changes what the system can intentionally do with its own conceptual structure*.

In Hyperseed terms: attention is the deliberate allocation of limited resources.[22] A system that can bring its own ontological commitments into the focus of attention can devote resources to understanding and revising them. A system whose ontological commitments are trapped in parts of its mind that do not interface cleanly with deliberative reasoning may still adapt, but without the same degree of transparent self-understanding and controlled self-modification.

A formal framing. Let L be the system's deliberative representation language (the language in which it formulates explicit hypotheses and plans). Let Θ be a space of internal parameters/structures that determine behavior (which may include subsymbolic encodings). Let $\mathcal{M}$ be a self-modification operator that proposes changes in Θ based on reasoning in L.

- In an *implicit-ontology* regime, large parts of the system's core conceptual commitments live in Θ but lack a faithful, manipulable representation in L.
- In an *explicit-ontology* regime, there is a reasonably faithful mapping from a substantial subset of those commitments into L (e.g. via explicit Hyperseed nodes/relations), so that reasoning processes in L can refer to, critique, and revise them.

This leads to a monotonicity-style advantage claim:

Proposition 4 (Explicit ontological access expands the reachable space of deliberate self-modification)**.** *Assume: (i) the system's self-modification proposals are generated by reasoning in L; (ii) the system can only deliberately modify aspects of Θ that are representable (with usable fidelity) in L; and (iii) the evaluation criterion for self-modification is monotone with respect to the set of available deliberate modifications (the system may ignore any proposal). Then, holding everything else fixed, moving from an implicit-ontology regime to an explicit-ontology regime (by adding an explicit core*

[22] Hyperseed-Concept 60 (*Attention*) and Hyperseed-Concept 100 (*Genenergy / Effort*).

ontology such as Hyperseed into L) weakly expands the set of reachable deliberate self-modifications and therefore cannot decrease the optimum achievable value of the evaluation criterion under a fixed resource budget.

Proof sketch. By assumption, the set of deliberate modifications available in the implicit regime is a subset of those available in the explicit regime, because explicit representability in L is a precondition for deliberate modification. Since the evaluation criterion is monotone in the available choice set, the optimum over the larger set is at least the optimum over the subset. □

Why this matters in practice. The formal claim is intentionally modest (a non-degradation statement), but it points to two practical AGI advantages of explicit ontology:

- **Better self-understanding.** The system can form explicit hypotheses about why it is confused, what distinctions it is failing to make, and what conceptual bridges it lacks—the same "distinction-first" thinking underlying Hyperseed's treatment of information and probabilistic belief.[23]
- **More controllable self-modification.** The system can propose and evaluate ontological revisions as explicit candidates—e.g. adding, removing, or restructuring concept links—with explicit resource/effort accounting.[24] When the ontology is only implicit, analogous changes may still occur via learning, but they may be harder to steer using deliberative reasoning.

In short: an explicit Hyperseed-like core ontology does not replace learning; it changes the interface between learning, reasoning, and reflection. It makes the ontology-directedness of experience explicit, and that explicitness supports more transparent attention allocation and more targeted expenditure of genenergy toward conceptual revision.[25]

7 Automated Ontology Learning: Growing and Refining the Scrutability Scaffold

We have so far been treating Hyperseed ontology as a given, and looking at how to leverage it for effective AGI-oriented reasoning. However, Hyperseed itself, in its current form, is best considered a crude first approximation of what an AGI system's ontology needs to be. Where things get really interesting is when once considers using Hyperseed as an initial condition for automated ontology learning.

Of course, one can also do automated ontology learning from scratch, setting Hyperseed aside. However, it seems likely that to do this really effectively might require dramatically more compute time and diversity of system experience than incrementally updating a reasonable initially-provided ontology such as Hyperseed.

The preceding sections give a clear foundation for automated ontology learning, via treating an ontology not only as a representational artifact but as an *inference-control scaffold*: a base vocabulary B that induces reduction profiles (Definition 4), which in turn guide search via expected marginal information gain per unit cost. The point is, once we can *measure* how well an ontology supports this role—e.g. via the Ontological Efficiency Ratio (OER) and the associated diagnostics

[23] Hyperseed-Concept 98 (*Information as Distinction*) and Hyperseed-Concept 139 (*Probabilistic Belief*).
[24] Hyperseed-Concept 100 (*Genenergy / Effort*).
[25] Hyperseed-Concept 60 (*Attention*) and Hyperseed-Concept 100 (*Genenergy / Effort*).

(Subsection 4.5) built on Theorem 4 and the separator view (Definition 9, Theorem 5)—it becomes natural to treat ontology design itself as a learnable optimization problem.

In this spirit, we now sketch two complementary algorithmic routes to *automated ontology learning* that fit the paper's formal framing:

- **Route A (grow from scratch):** induce a compact base and derived concept definitions from inference histories, treating concept invention as a compression and separation problem.
- **Route B (refine an existing ontology):** start from a given scaffold (e.g. Hyperseed) and iteratively repair blind spots and add missing mediators, guided by the same OER/separation diagnostics.

Both routes assume the ontology is more than a term hierarchy: it is a collection of concepts specified by logical expressions (possibly in a higher-order, paraconsistent, or type-theoretic setting), plus bridge axioms that connect abstract predicates to domain-anchored ones.

7.1 Ontology learning as optimization over definitional theories

Fix a background logic/language $\mathcal{L}$ used by the reasoner. (For PLN+Hyperon, $\mathcal{L}$ is the uncertain logic supported by the Atomspace representation; for other implementations, $\mathcal{L}$ could be a paraconsistent predicate logic or a typed logic.) We treat an ontology instance as a pair

$$\mathcal{O} = (B, \Delta),$$

where B is the designated base vocabulary (the "scrutability primitives" used for profiles) and Δ is a set of definitional and bridge constraints.

A single learned concept C is represented not merely by a node label but by a *definitional schema* in $\mathcal{L}$, e.g.

$$\forall x \;\; C(x) \leftrightarrow \varphi_C(x)$$

or (in an uncertain / non-brittle encoding) by a pair of implications with graded support,

$$C(x) \Rightarrow \varphi_C(x), \qquad \varphi_C(x) \Rightarrow C(x),$$

each carrying truth-value annotations appropriate to the reasoner. The intent is that φ_C uses previously available symbols (and possibly second-order constructors) so that adding C constitutes a genuine abstraction/compression rather than a trivializing encoding (cf. Theorem 2).

Data: inference histories and task episodes. Let $\mathcal{D}$ be a collection of reasoning episodes (queries and tasks), and let $\mathcal{H}(\mathcal{O})$ denote the inference histories produced by running the system under ontology $\mathcal{O}$ (including rule applications, premise bindings, costs, and derived conclusions). These traces support:

- estimating reduction-quality signals (mutual information and residual uncertainty; Section 2.4);
- estimating control-quality signals (OER, $\hat{\tau}$, $q(v)$ tails, residual component structure; Subsection 4.5);
- proposing candidate abstractions by mining repeated proof fragments and repeated cross-domain bridge patterns.

A generic objective. We will not commit to a single scalar objective, but a representative loss functional is

$$\mathcal{L}(\mathcal{O}) \;=\; \lambda_{\text{scrut}} \cdot \text{ScrutLoss}(\mathcal{D}, \mathcal{O}) \;+\; \lambda_{\text{ctrl}} \cdot \widehat{\text{OER}}(B) \;+\; \lambda_{\text{sep}} \cdot \text{Leak}(B) \;+\; \lambda_{\text{comp}} \cdot \text{Comp}(\mathcal{O}), \quad (4)$$

where:

- ScrutLoss measures how well target concepts become ϵ-scrutable from B (Definition 2 and Definition 4), e.g. via residual uncertainty or held-out prediction error of reduction profiles;
- $\widehat{\text{OER}}(B)$ is estimated from traces as in Subsection 4.5;
- $\text{Leak}(B)$ is the leakage term from AOS (Definition 9);
- $\text{Comp}(\mathcal{O})$ is an MDL-style proxy for "how much ontology we added," approximating the complexity regularization principles in Theorem 2 (e.g. syntactic description length of Δ plus $|B|$ or proxy Kolmogorov complexity estimates).

The key engineering point is that *ontology learning is driven by the same quantities that control inference*: information gain and effort, together with structural separation signals that predict exponential speedup.

7.2 Route A: growing an ontology from scratch from inference histories

Route A assumes we do not begin with a curated base like Hyperseed, but instead start from a minimal symbolic substrate and a stream of inference histories. The goal is to *invent* a compact base B and definitional theory Δ that make broad families of concepts increasingly scrutable and make inference controllable.

A.1 Candidate-generation by compression of proof fragments. A practical way to invent predicates is to mine repeated structure in inference traces. Two common trace-level motifs are:

- **Repeated conjunctive/relational motifs:** the same small pattern of premises repeatedly appears as the "useful core" of a subproof. Introduce a new predicate C whose definition φ_C captures that motif, so later proofs can use C as a single handle.
- **Repeated bridge motifs:** similar cross-domain moves repeatedly occur, but only after long detours. Introduce a mediator predicate M whose definition makes the bridge explicit, so the concept relevance graph becomes more nearly separated by a thin waist (AOS), reducing $\bar{q}$ and thus OER.

In both cases, the candidate space of φ_C can be generated by a grammar of allowed constructors in $\mathcal{L}$ (conjunction, implication chains, bounded quantification over roles/relations, higher-order pattern constructors if available, etc.). Candidates are then filtered by whether they (i) occur frequently enough to pay for their representational cost, and (ii) improve separation or mutual-information coverage.

A.2 Base-set selection as separator learning. Even with many derived predicates available, the designated base B matters because it is the feature set used for reduction profiles and for cluster ranking. Using the concept relevance graph G_η (Definition 8), one can search for B that approximately separates the graph (Definition 9) while keeping $|B|$ small. Operationally, this is guided by the diagnostic loop: progressively add candidate base predicates and watch $\widehat{\text{OER}}(n)$ plateau (Subsection 4.5).

A.3 Trace-driven learning loop (sketch). A minimal end-to-end Route A loop is:

```
# Route A: GrowOntologyFromScratch
initialize ontology O = (B, Delta) with a minimal seed vocabulary
repeat until budget exhausted:
  run reasoner on tasks D using O, collect inference traces H
  estimate diagnostics from H:
     OER_hat, tau_hat, residual_components, leakage_hat, etc.
  mine candidate concept definitions Phi from H:
     - frequent proof-fragment motifs
     - frequent cross-domain bridge motifs
  for each candidate definition phi in Phi:
     form edit e = "add new predicate C with definition C <-> phi"
     score[e] = predicted_improvement(OER_hat, leakage_hat, scrut_loss)
                - complexity_penalty(e)
  apply best-scoring edit(s) to O
  occasionally:
     - prune redundant concepts (low use / low MI gain)
     - re-select base set B to improve separator quality
return O
```

This loop is deliberately conservative: it uses the same "pay rent" criterion as inference control itself. A new concept is retained only if it reduces expected scrutability loss and/or improves separator structure enough to decrease OER, after penalizing the additional descriptive complexity.

7.3 Route B: incremental refinement of a given ontology (e.g. Hyperseed)

Route B begins from an existing curated scaffold $\mathcal{O}_0 = (B_0, \Delta_0)$ and treats ontology learning as *debugging and enrichment* rather than open-ended invention. This setting is directly motivated by the paper's diagnostic program (Subsection 4.5), which already identifies where an ontology fails as an inference-control scaffold.

B.1 Where to edit: using the diagnostic signals. The following signals naturally suggest specific ontology edits:

- **High OCS violation rate $\hat{\tau}$:** systematic misranking indicates missing mediator concepts or missing bridge axioms in Δ that would make the ontological projection π_B track value-relevant structure more faithfully.
- **Heavy tails in $q(v)$:** there are inference contexts in which the ontology does not discriminate. These contexts indicate candidate areas for enrichment: add a small number of concepts whose definitions carve that context into better-separated subcases.
- **Large residual components under AOS:** if removing B leaves large components, the ontology is not acting as a thin waist for those domains. Add concepts whose definitions are built from patterns common across those domains (abstractions), or add bridge axioms that force more dependencies to flow through existing ontology nodes.
- **Plateau behavior of $\widehat{\mathrm{OER}}(n)$:** if OER does not improve as $|B|$ grows, the candidate additions are likely either redundant or overly concrete. This suggests changing the *kind* of

abstraction being introduced (i.e. different grammar/operators for φ_C), not merely adding more terms.

B.2 A small set of ontology edit operators. A practical refinement system can be built from a small menu of local edit operators:

- **Add a new mediator concept:** introduce C with a definition $C \leftrightarrow \varphi_C$ intended to reduce leakage between two residual regions.
- **Split an overloaded concept:** replace C by C_1, C_2 with definitions that partition its typical uses, when traces show that a single predicate is used in two semantically distant ways that confuse control.
- **Merge redundant concepts:** if C_1 and C_2 are nearly equivalent across traces (high mutual information / near-implication in both directions), merge to reduce complexity without losing control quality.
- **Repair bridges:** add, remove, or re-weight bridge axioms linking abstract concepts to domain-anchored predicates, guided by whether those bridges actually reduce scrutability loss or improve control.

B.3 Trace-driven refinement loop (sketch). A minimal Route B loop is the same skeleton as Route A but with a more targeted proposal step:

```
# Route B: RefineOntology
input: initial ontology O0 = (B0, Delta0)
O = O0
repeat until budget exhausted:
  run reasoner on tasks D using O, collect traces H
  compute diagnostics: OER_hat, tau_hat, q_tail, residual_components, etc.
  identify "bad contexts" where the ontology provides weak discrimination
  propose a small set of edits E:
     - add mediator concepts targeting largest residual components / highest leakage
     - split concepts implicated in high tau_hat contexts
     - repair or add bridge axioms where grounding fails
  for each edit e in E:
     estimate delta_score[e] on held-out traces or via fast surrogate evaluation
     score[e] = delta_score[e] - complexity_penalty(e)
  apply best-scoring edit(s), update base set B if needed
return O
```

The key difference from Route A is that refinement is constrained by a strong prior structure (the existing ontology), so edits should be local and justifiable in terms of the diagnostics. This aligns with the engineering stance that an ontology should be *legible* and *maintainable*: growth without repair discipline risks creating a large, fragile concept soup that does not function as a clean separator.

7.4 Remarks on richer logics and mechanization

Nothing in the above learning loops depends on a specific choice of $\mathcal{L}$ beyond three requirements:

1. **Evaluability:** we must be able to evaluate candidate definitions (or at least estimate their predictive impact) on traces, so that mutual-information and scrutability-style objectives can be computed.
2. **Compositional cost:** we must be able to assign an effort/complexity proxy to candidate formulas, so that trivializing encodings are disfavored in the spirit of Theorem 2.
3. **Operational linkage to control:** the ontology must induce the projection π_B used by inference control, so that improvements in separation and reduction quality translate into improvements in search behavior.

Thus the same overall approach applies whether concepts are expressed in a higher-order predicate logic, a paraconsistent logic with explicit support and counter-support channels, or a type-theoretic setting in which concept definitions are types/terms rather than predicates. The unifying theme is that an ontology is a *learned, compression-oriented interface* between raw domain detail and the thin waist required for scalable inference control.

8 Discussion: Practicalities, Limitations, and Research Directions

The material above deliberately spans two genres: philosophy of concepts (scrutability and reduction) and engineering (inference control). This section sketches how the two constrain each other once the system is built and deployed.

8.1 Operational meaning of "approximate reduction"

In a PLN+Hyperseed implementation, "reducing a concept W to Hyperseed" is not primarily a claim about definitional equivalence. It is an algorithmic act: build a reduction profile (or multiple competing profiles) that makes W more predictable, more compressible, and more inferentially connected to the rest of the knowledge base. The reduction may improve over time, and multiple reductions may coexist, indexed by contexts and aspects. This is the intended engineering analogue of a scrutability-base program: not a final, perfect reduction, but a systematic process that yields increasingly usable analyses.

8.2 Inference control is where the philosophy pays rent

In large uncertain reasoning systems, the primary bottleneck is rarely the availability of inference rules; it is the combinatorial explosion of possible rule applications and premise bindings. The Hyperseed-guided view suggests that a concept reduction profile is not merely explanatory; it is a feature vector for control. Once an active query is embedded into Hyperseed space, the system can prioritize steps that are (i) close in the ontology, (ii) high in expected information gain, and (iii) low in estimated cost.

8.3 Embodiment and domain anchoring are not optional

A common failure mode of purely abstract ontologies is ungrounded inference: beautiful chains of symbolic steps that do not connect to the world, the body, or the domain of concern. The bridging proposition (Proposition 1) formalizes a minimal engineering stance: adding sound Hyperseed–domain bridges can only help if the controller is allowed to ignore them and if evaluation is monotone in available proof paths. The hard work is building bridges that are (i) correct enough, (ii) rich enough to matter, and (iii) cheap enough to be used frequently under budget constraints.

8.4 Three complementary learning loops

The overall picture suggests three coupled learning loops:

- **Reduction learning:** learn better Hyperseed reduction profiles for concepts as the system accumulates evidence and experience—a "semantic compression" loop.
- **Control learning:** learn better priors over rule schemas and better relevance estimators for premise selection (including association-spreading dynamics)—an "attention allocation" loop.
- **Ontology learning:** learn to revise the base vocabulary B and the definitional/bridge theory Δ itself (concept invention, splitting/merging, and targeted bridge construction), using the trace-based diagnostics of Subsection 4.5 (OER, $\hat{\tau}$, $q(v)$ tails, residual component analysis) as direct optimization signals—a "scaffold revision" loop (see Section 7).

The philosophical scrutability view gives a normative target for the reduction and learning loops; the engineering constraints of inference control give a normative target for the control loop.

9 Conclusion

We have articulated a unified argument—philosophical in motivation and engineering in consequence—for treating Hyperseed as a practical cousin of a Chalmers-style scrutability base, realized not by crisp definitions but by uncertain, intensional reductions built and refined through PLN inference. The core philosophical move is to replace brittle definitional reduction with probabilistic, profile-based scrutability: concepts become increasingly scrutable from a compact base as their Hyperseed reduction profiles improve and as the system learns which primitives provide the largest marginal information gain.

The core engineering move is to use those same reductions as inference-control scaffolding: backward and forward chaining become tractable when guided by marginal information gain per unit cost, and bidirectional geodesic search provides a principled way to merge premise-side reachability with goal-side pull while maintaining explicit resource accounting and (in richer implementations) evidence conservation.

Finally, we argued for an AGI-oriented meta-point: making a core ontology explicit in a system's deliberative language expands the reachable space of deliberate self-understanding and self-modification. Implicit ontologies will emerge in any sufficiently capable learning system, but explicit ontologies make ontological commitments legible to the system itself, turning them into objects of reflection and controlled revision. In the Hyperseed+PLN picture, this is not merely a philosophical preference; it is an architectural affordance that can be used to guide inference, learning, and reflective cognition under bounded resources.

References

[1] Rudolf Carnap. *The Logical Structure of the World (Der logische Aufbau der Welt)*. Open Court, 1967 (original German edition 1928).

[2] Anna Wierzbicka. *Semantics, Culture, and Cognition: Universal Human Concepts in Culture-Specific Configurations.* Oxford University Press, 1992.

[3] George A. Miller. WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41, 1995.

[4] Charles J. Fillmore, Christopher R. Johnson, and Miriam R. L. Petruck. Background to FrameNet. *International Journal of Lexicography*, 16(3):235–250, 2003.

[5] David J. Chalmers. *Constructing the World.* Oxford University Press, 2012.

[6] Ben Goertzel et al. Hyperon: A metacognitive, neural-symbolic AGI framework (review/overview paper). arXiv preprint (exact identifier to be filled).

[7] Ben Goertzel, Matt Iklé, Izabela Goertzel, and Ari Heljakka. *Probabilistic Logic Networks: A Comprehensive Framework for Uncertain Inference.* Springer, 2008.

[8] Ben Goertzel. Hyperseed ontology: formal presentation and theoretical development. Unpublished manuscript, January 2026.

[9] Ben Goertzel and ChatGPT 5.2-Pro. Motivated wu-wei creativity: a Hyperseed argument for intention-coupled generative AI. Position paper, March 2026.